

# Cross-Family Symbolic Verification for Contamination-Robust Selective Prediction on LLM Math Reasoning

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## Abstract

Selective prediction for LLM math reasoning fails along two coupled axes: same-model self-verification suffers correlated-error collapse (13.8% false acceptance on MATH-500, scaling monotonically from 1.4% at Level 1 to 30.8% at Level 5), and agreement-based signals look strong on MATH-500 but collapse on benchmarks released after the solver’s training cutoff. We introduce X-SGRV, in which a large LLM from a different family than the solver reads only the problem statement and emits a SymPy `verify(answer)` function executed in a sandboxed subprocess; a gold-free adversarial filter and cross-extractor consensus eliminate residual false positives. On the pre-registered full MATH-500 ( $n=498$ ), strict two-way consensus accepts 171 candidates and is correct on all 171 (95% Clopper-Pearson CI  $[0.979, 1.000]$ ) at 34.3% coverage. On contamination-clean AIME 2025 and a 125-problem post-cutoff combo, coverage correctly collapses to 3–12% while accepted answers remain right. At matched coverage X-SGRV ties Skywork-o1-Open-PRM and doubles Qwen-Math-PRM on the contamination-clean combo.

## 1. Introduction

Selective prediction on LLM math reasoning needs a deployment-time signal indicating which answers should be trusted. Two coupled failure modes obstruct this. First, same-model self-verification is correlated by construction: when the solver also writes the assertions checking its work, errors in the solution propagate into the assertions, and the verifier inher-

its the solver’s blind spots (Huang et al., 2024; Kim et al., 2025; Stechly et al., 2024). Second, MATH-500 is partially memorized by the standard solvers used to study it: Qwen2.5-Math-7B reproduces 54.6% of the benchmark verbatim from a 60% prefix, with accuracy dropping sharply on post-cutoff problems (Wu et al., 2025). High top-tier precision on MATH-500 therefore confounds verification reliability with memorization.

Three prior lines address one failure but not both. Code-based same-model self-verification (Wang et al., 2024a;b) is executable but does not break the solver/verifier coupling. Process reward models (Lightman et al., 2024; Zhang et al., 2025; Skywork Team, 2024) inherit the contamination footprint of their training distribution. Autoformalization checks (Zhou et al., 2024) achieve structural independence but require heavyweight formal-methods infrastructure with reported coverage below 20% on competition problems.

**Contribution.** We introduce X-SGRV (Figure 1), in which a large LLM from a different model family than the solver reads only the problem statement and emits a Python `verify(answer)` function executed in a SymPy sandbox. A gold-free deployment-time adversarial filter probes the verifier against perturbations of the candidate; cross-extractor consensus requires multiple independent extractors to all produce working verifiers and all accept. On the pre-registered full MATH-500, strict two-way consensus is correct on 171/171 accepts ( $[0.979, 1.000]$ ) at 34.3% coverage; on contamination-clean benchmarks the system correctly collapses to single-digit coverage while preserving precision; at matched coverage it ties Skywork-PRM and doubles Qwen-PRM on the contamination-clean combo, while requiring no GPU infrastructure.

## 2. Method

A solver  $M$  produces a plurality candidate  $\hat{a}$  for problem  $x$ . A cross-family extractor  $E$ , drawn from a model family architecturally distinct from  $M$ , receives only  $x$  and emits a Python function `verify(answer)` -

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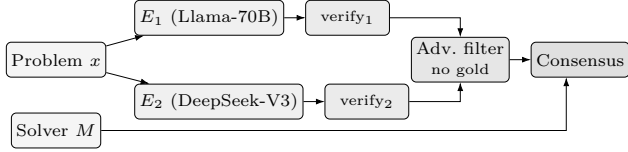


Figure 1. X-SGRV pipeline. Independent cross-family extractors  $E_1, E_2$  read the problem statement only and emit SymPy verify functions. The deployment-time filter probes each verifier against perturbations  $\{\hat{a} \pm 1, \hat{a} + 7, \hat{a} \times 2\}$  of the solver’s candidate  $\hat{a}$ ; verifiers accepting any probe are marked broken. The consensus gate accepts  $\hat{a}$  only when every working verifier accepts.

> bool that re-derives the problem’s constraints from the statement in SymPy plus standard library primitives. The prompt forbids hard-coding the answer and requires the literal token UNVERIFIABLE when no bounded check is possible.

The emitted script is executed in a Python subprocess sandbox with a 10-second SIGALRM timeout. The deployment-time adversarial filter probes each verifier against  $\mathcal{P}(\hat{a}) = \{\hat{a} - 1, \hat{a} + 1, \hat{a} + 7, \hat{a} \times 2\}$  (for integer  $\hat{a}$ ; deterministic fallback for non-integers). A verifier that accepts any probe is marked broken. Strict  $k$ -way cross-extractor consensus accepts  $\hat{a}$  only when every participating extractor produces a working verifier and every working verifier accepts. The probe set and consensus rule were locked in a pre-registration document committed before the  $n=498$  scale-up. Implementation details, full extractor prompt, baseline configurations, and statistical methodology are in ????

### 3. Experiments

Setup. Solver: Qwen2.5-7B-Instruct-Turbo (Together AI). Extractors: Llama-3.3-70B, DeepSeek-V3, gpt-oss-120b, Claude Sonnet 4.6, Qwen3-Coder-480B. Benchmarks: full MATH-500 ( $n=498$  after solver timeouts), stratified MATH-500 ( $n=175$ , balanced across Levels 1–5), AIME 2025 ( $n=30$ ), and a CleanMath combo of HMMT-Feb/BRUMO/SMT/APEX 2025 ( $n=125$ ). Baselines: 4/10-sample self-consistency (Wang et al., 2023), semantic entropy (Kuhn et al., 2023; Farquhar et al., 2024),  $p(\text{True})$  (Kadavath et al., 2022), Qwen2.5-Math-PRM-7B (Zhang et al., 2025) and Skywork-o1-Open-PRM-7B (Skywork Team, 2024) run on Modal A100; PRMs validated against ProcessBench (Zheng et al., 2024).

Same-model FAR scales with difficulty. Same-model SymPy self-verification accepts 320 MATH-500 candidates with a 13.8% false acceptance rate, scaling monotonically from 1.4% at Level 1 to 30.8% at Level 5; a  $2 \times 2$  ablation isolates solver-verifier independence,

not test-input randomization, as the operative axis (Fisher’s exact  $p < 10^{-8}$ ; ??).

Single-extractor X-SGRV. On the full MATH-500 ( $n=498$ ), a Llama-3.3-70B extractor accepts 243 candidates and is correct on 230, precision 0.947 ([0.91, 0.97]) at 48.8% coverage with zero gold-relative adversarial false positives over 964 probes. A DeepSeek-V3 extractor reaches  $199/207 = 0.961$  ([0.93, 0.98]) at 41.6% coverage.

Cross-extractor consensus drives residual FPs to zero (Table 1). Strict two-way consensus ( $\text{Llama} \cap \text{DeepSeek-V3}$ ) on the pre-registered full MATH-500 accepts 171 candidates and is correct on all 171 (95% CI [0.979, 1.000]) at 34.3% coverage. Adding three further extractors — gpt-oss-120b, Claude Sonnet 4.6, Qwen3-Coder-480B — to form a five-way consensus on stratified MATH-175 yields  $69/69 = 1.000$  ([0.948, 1.000]). Adding a sixth extractor (GPT-5-mini) reduces the accepted tier to  $67/67 = 1.000$  ([0.946, 1.000]). Adding extractors monotonically shrinks the accepted tier while preserving perfect precision.

Coverage correctly collapses on contamination-clean benchmarks. On AIME 2025 the deployment-time filter removes the single broken verifier and yields  $1/1 = 1.000$  at 3.3% coverage; on CleanMath the system abstains on 121/125 problems and accepts 4 correct candidates (precision 1.000). A rotated solver (DeepSeek-V3.1) raises the CleanMath accept count to 15/15, isolating the verifier’s selectivity from solver accuracy.

Matched-coverage comparison against PRMs (Table 2). At 54.9% coverage on MATH-175, X-SGRV, Qwen-PRM (min-step), and Skywork-PRM (product-step) all achieve precision 0.927–0.938; on the contamination-clean CleanMath at 3.2% coverage, X-SGRV and Skywork-PRM (mean-step) reach  $4/4 = 1.000$  while Qwen-PRM (min-step) reaches  $2/4 = 0.500$ , a split consistent with Qwen-PRM’s training distribution being more concentrated on MATH-family problems.

### 4. Discussion

The 22-fold scaling of same-model false acceptance from L1 to L5 is consistent with correlated-error collapse: when one model produces both the solution and the assertions checking it, the assertions describe the model’s misreading of the problem (Kim et al., 2025). Specification grounding, by deriving the check from the problem statement, breaks this overlap. The  $2 \times 2$  ablation shows the operative axis is solver-verifier independence, not test-input randomization (??).

Table 1. X-SGRV top-tier precision across extractors, benchmarks, and consensus configurations. “Cov.” is the size of the top tier; “Prec.” is 95% Clopper-Pearson. CleanMath = HMMT-Feb + BRUMO + SMT + APEX 2025 ( $n=125$ ).

| Extractor                     | Benchmark          | Cov.  | Prec. (95% CI)  |
|-------------------------------|--------------------|-------|-----------------|
| Llama-3.3-70B                 | MATH-500           | 48.8% | 230/243 = 0.947 |
| DeepSeek-V3                   | MATH-500           | 41.6% | 199/207 = 0.961 |
| Claude Sonnet 4.6             | AIME 2025          | 10.0% | 3/3 = 1.000     |
| Llama-3.3-70B                 | CleanMath          | 3.2%  | 4/4 = 1.000     |
| 2-way consensus               | MATH-500           | 34.3% | 171/171 = 1.000 |
| 5-way consensus               | MATH-175           | 39.4% | 69/69 = 1.000   |
| Llama (Solver: DeepSeek-V3.1) | CleanMath          | 12.0% | 15/15 = 1.000   |
| Llama-3.3-70B                 | Omni-MATH hard-100 | 1.0%  | 0/1 = 0.000     |

Table 2. X-SGRV versus open PRMs at matched coverage.

| Method              | Benchmark | Cov.  | Prec. (95% CI)     |
|---------------------|-----------|-------|--------------------|
| X-SGRV              | MATH-175  | 54.9% | 0.938 [0.87, 0.98] |
| Qwen-PRM (min)      | MATH-175  | 54.9% | 0.927 [0.86, 0.97] |
| Skywork-PRM (prod.) | MATH-175  | 54.9% | 0.938 [0.87, 0.98] |
| X-SGRV              | CleanMath | 3.2%  | 1.000 [0.40, 1.00] |
| Qwen-PRM (min)      | CleanMath | 3.2%  | 0.500 [0.07, 0.93] |
| Skywork-PRM (mean)  | CleanMath | 3.2%  | 1.000 [0.40, 1.00] |

Under approximate independence between cross-family extractors, the joint false-positive rate is bounded above by the product of per-extractor rates: empirically  $0.053 \times 0.039 \approx 0.002$ , consistent with the observed 0/171. The mechanism is lab-agnostic: five extractors drawn from Meta, DeepSeek, OpenAI, Anthropic, and Alibaba agree on 69/69 accepts.

X-SGRV is not a continuous-score selective-prediction signal. Top-tier coverage on contamination-clean benchmarks is single-digit by design; on the hardest 100 Omni-MATH problems zero working verifiers are produced, and on PutnamBench-style proof problems the extractor self-detects scope by emitting UNVERIFIABLE on roughly half of a 100-problem sample. The intended deployment posture is escalation, not commitment, when the verifier abstains. Three concrete falsifiable follow-ups: a contamination-stratified replication with per-problem prefix-completion overlap; an extractor-scaling sweep at 13B/32B/70B; and a hybrid SymPy-plus-Lean pipeline (78.9% Lean compile rate observed for Claude Sonnet 4.6 against Mathlib).

## Reproducibility statement

All models are public via Together AI, OpenAI, Anthropic, or HuggingFace. The X-SGRV implementation, sandbox harness, adversarial filter, full extractor prompt, result JSONs, and pre-registration document are provided in the anonymized supplementary code release.

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